

Project Report: GenAI Image Dataset and Video Output

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1 Introduction

This project focuses on the creation and utilization of a generative AI-based image dataset, with the final output being a single video. The video simulates a discussion between two or more personalities, leveraging images and/or video data from the curated dataset. The work aligns with the guidelines discussed in class, emphasizing dataset preparation, model selection, and multimedia output generation.

2 Work Performed

The core component of this project is the development and execution of a robust Python script, `genai_image_dataset.py`, which automates the process of large-scale image dataset creation. This script caters to various categories such as *cat, dog, car, airplane, flower, tree, building, person, food, and bicycle*.

2.1 Script Design and Implementation

The script is designed with modularity, reliability, and scalability in mind. Its key features and workflow are as follows:

- **Parameterization:** The script defines configurable parameters, including the list of target categories, the dataset directory structure, the number of images to be collected per category (2,000), and a minimum image size threshold (50 KB) to ensure quality.
- **Directory Management:** For each category, the script programmatically creates a dedicated subdirectory within the main dataset folder, ensuring organized storage and easy access for downstream tasks.
- **Image Retrieval:** Leveraging the DuckDuckGo search engine via the DDGS Python library, the script performs keyword-based image searches for each category. It requests a surplus of image results to account for potential filtering and download failures.
- **Download and Validation:** For each retrieved image URL, the script attempts to download the image using the Requests library. It validates each image by checking the HTTP response status and enforcing the minimum file size requirement, thereby filtering out low-quality or irrelevant images.

- **File Naming and Storage:** Images are saved with a standardized, zero-padded naming convention (e.g., 00000.jpg), which facilitates dataset integrity and simplifies future batch processing or annotation tasks.
- **Error Handling and Robustness:** The script incorporates comprehensive error handling, including exception management for network issues and invalid URLs. It gracefully skips failed downloads and continues the process until the target number of valid images is reached for each category.
- **Progress Monitoring:** The integration of the tqdm library provides real-time progress bars for each category, enhancing user experience and transparency during dataset generation.

2.2 Expected Output

Upon successful execution, the script produces a highly organized dataset directory containing up to 2,000 high-quality images for each specified category. Each category's images are stored in their respective subfolders, named accordingly. This dataset is well-suited for a variety of machine learning and computer vision applications, including generative AI, image classification, and object detection.

2.3 Submission Details

The submitted zip file includes both the generated image dataset and the `genai_image_data`.

3 Expected Output

- **Primary Output:** A single video file (.mp4 or .avi) showcasing a discussion between AI-generated personalities, utilizing images from the dataset.
- **Dataset:** A well-organized image dataset, ready for further AI or multimedia projects.

4 Submission Details

The entire project, including the video output, image dataset, and this report, is submitted as a zip file. The zip file contains:

- The final video showcasing the discussion between AI-generated personalities.
- The organized image dataset used for the project.
- This detailed report documenting the work performed, expected output, and model used.

5 Conclusion

This project demonstrates the end-to-end workflow of dataset preparation, generative AI model usage, and multimedia output creation.